

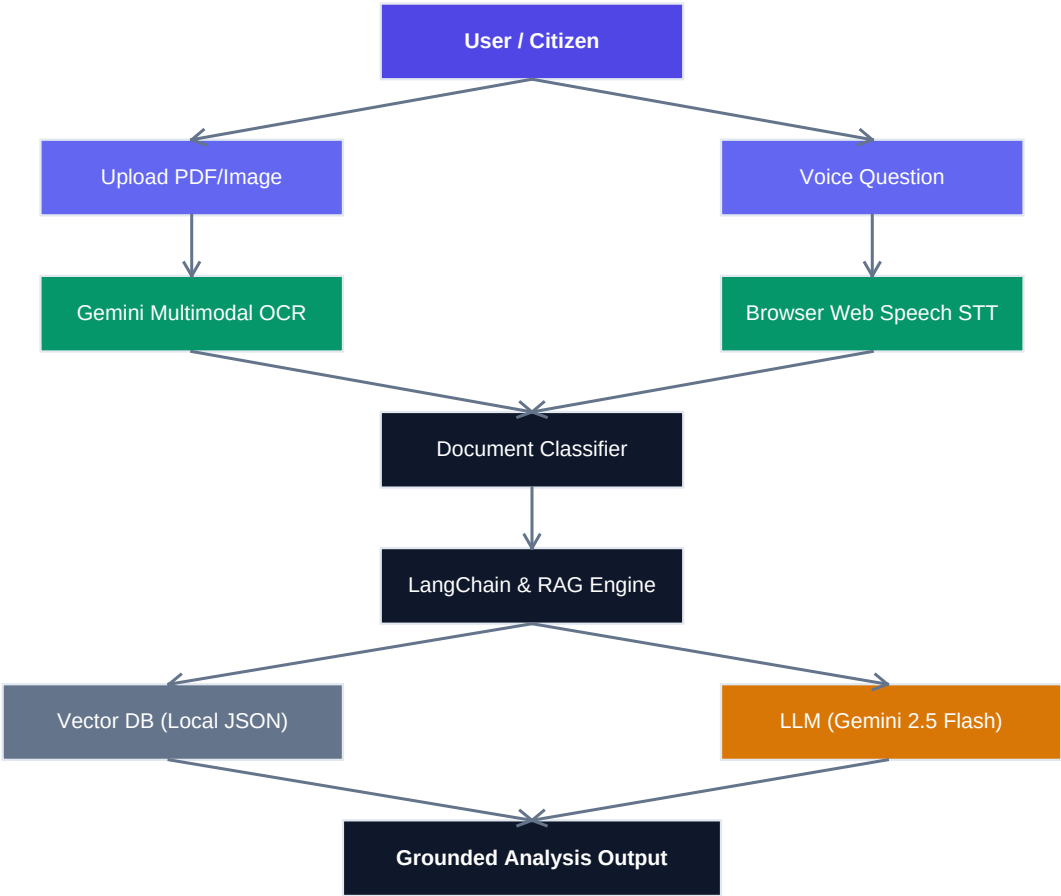
# NyayaMitra AI

## System Architecture and Application Flow Map

Date: July 26, 2026 | Author: Project Team & Antigravity AI

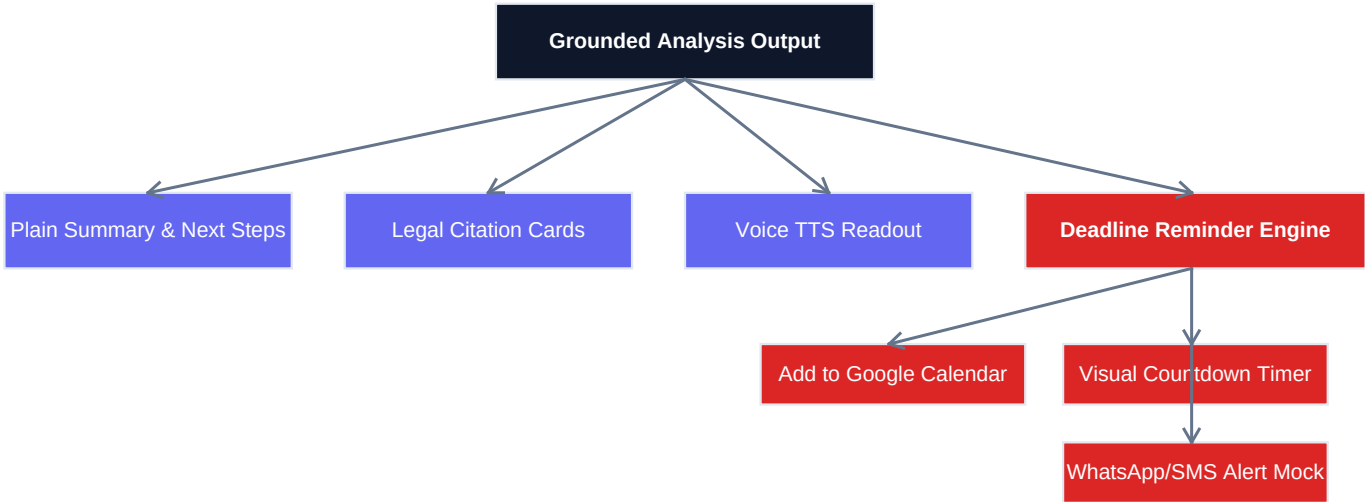
### 1. Application Flow and Architecture Map

Below is the structural map of the NyayaMitra AI prototype. It outlines how user inputs flow from ingestion and native OCR / Speech-to-Text conversion into the Document Classifier and LangChain RAG pipeline, generating simplified outputs and automated calendar/SMS reminders.



## 2. Synthesis and Output Channels

Once the backend generates the grounded analysis, the system splits output processing into two parallel flows: UI display cards and the Deadline Alert / Reminder queue.



### Deep-Dive on Key Architecture Stages

#### [1] A. Ingestion and Native OCR

- Unlike standard setups that deploy local OCR systems like Tesseract or PaddleOCR (which require heavy compilation), NyayaMitra utilizes Gemini's native multimodal vision. This results in far higher accuracy for complex legal layouts.

#### [2] B. LangChain and RAG Engine

- A custom Legal Knowledge Base (JSON format) stores curated summaries of key Indian Civil/Notice laws. The RAG engine runs keyword and embedding similarity searches to extract relevant citations before invoking the generator.

#### [3] C. Grounded Analysis Output

- The system produces structured JSON data containing (1) Plain language summaries, (2) Citations, (3) An actionable next-step checklist, and (4) Formatted response templates.

#### [4] D. Voice Output and Translation Toggle

- Supports on-the-fly English to Telugu translation. Web Speech API synthesis allows text to be read aloud directly, ensuring digital inclusion for illiterate citizens.

### 3. Deadline Reminder Integration Details

Missing key legal deadlines can lead to default judgments, penalties, or loss of legal rights. The NyayaMitra AI Deadline Reminder Engine addresses this critical problem:

#### How the Deadline Pipeline Works

- **Date Parsing:** The LLM extracts date mentions from the notice text (e.g. 'within 15 days', 'appear on 2026-08-10') and parses them into ISO 8601 format.
- **Urgency Calculation:** Calculates the remaining time relative to the current local date and formats a warning level (High/Medium/Low).
- **Calendar Sync:** Generates an .ics file dynamically on the client-side, allowing users to sync the event with Google Calendar, iCal, or Outlook in one click.
- **Visual Alerting:** The frontend renders a prominent banner and vertical timeline displaying days remaining.
- **SMS/WhatsApp Scheduler:** Demonstrates a Twilio API mockup that schedules SMS or WhatsApp alerts to ping the citizen at 7-day, 3-day, and 24-hour milestones.

### 4. Tech Stack Optimization for Python 3.14.2 and Windows

To guarantee that the hackathon prototype runs successfully without installation issues, we have optimized the stack:

| Layer            | Technology Choice             | Reasoning & Optimization                                                       |
|------------------|-------------------------------|--------------------------------------------------------------------------------|
| Frontend         | React (Vite + TypeScript)     | Fast UI creation, smooth custom transitions and card layouts.                  |
| Backend          | FastAPI (Python 3.14.2)       | Asynchronous, fast execution, autogenerated interactive docs.                  |
| AI & OCR         | Gemini 2.5 Flash API          | Native OCR for PDFs and images. Eliminates local PaddleOCR compile errors.     |
| RAG Database     | Local JSON + Numpy Similarity | Replaces ChromaDB/FAISS. Pure-Python similarity prevents Windows build issues. |
| Voice Synthesis  | Web Speech API (Browser)      | Zero-cost, fast client-side audio rendering, native language support.          |
| Alerts/Reminders | Dynamic .ics + Twilio Mock    | Immediate integration with standard phone calendars; demo-ready alerts.        |

### 5. Core Hackathon Selling Points (IEEE Theme)

- **Digital Inclusion:** Acts as a bridge for the digitally excluded, using translation and audio readout.
- **Clarity First:** A clear, color-coded visual hierarchy reduces anxiety and helps users understand what to do.
- **Actionable Support:** Providing structured, download-ready legal templates lowers initial barriers to justice.